

day / date

mode > median > mean mean > median > mode.

Q1:

a) Skewness: A distribution in which the values ~~asymmetrical~~ from the mean have equal frequencies is defined in to symmetrical from any departure from symmetry.

b) Bowley's coeff. of skewness = $\frac{Q_3 + Q_3 - 2 \text{ Median}}{Q_3 - Q_1}$

$$Q_1 = L + \frac{h}{f} \left(\frac{n}{4} - c \right) \quad (Sc), \quad Q_3 = L + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$\text{Median} = L + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

→ highest freq.

$$f = 208, L = 24.5, c = 205, h = 5, n = 726.$$

29 → 29
176 → 205
208 → 413.
173

Q2:

a) Conditional Probability: It is the likelihood of an event occurring given that another event has already occurred. Denoted as

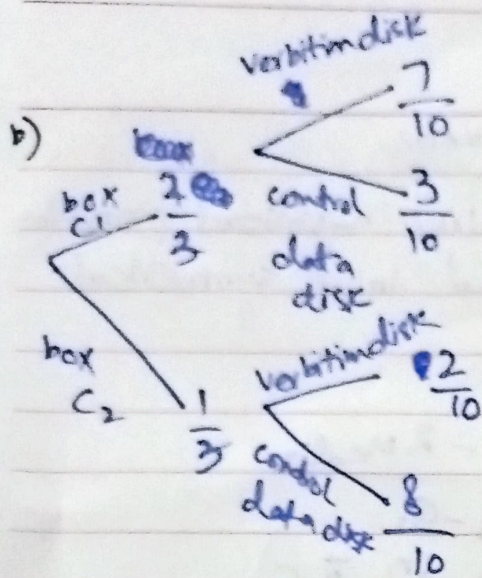
$P(A|B)$

, formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Prob. of A given that B.

Simple Events: An event that cannot be broken down into simpler components. Consists of a single outcome in sample space. e.g. when rolling a six-sided die, getting 3 is a simple event.

Compound Events: Consists of 2 or more simple events. Involves combination of multiple outcomes, e.g. when rolling a six-sided die, getting an even no (2, 4, 6) is a compound event b/c it combines multiple simple events.



1 box selected.
 $P(C_1) = \frac{2}{3}$, $P(C_2) = \frac{1}{3}$.

1 disk selected
 let C be disk from control/data disk.

$P(C_1|C) = ?$, $P(C_2|C) = ?$

$$P(C_1|C) = \frac{P(C_1 \text{ and } C) \times P(C_1)}{P(C)} \quad , \quad P(C_2|C) = \frac{P(C_2 \text{ and } C) \times P(C_2)}{P(C)}$$

~~scribbles~~

$$P(C_1|C) = \frac{\left(\frac{2}{3} \times \frac{3}{10}\right)}{\left(\frac{2}{3} \times \frac{3}{10}\right) + \left(\frac{1}{3} \times \frac{8}{10}\right)}$$

$$P(C_1|C) = \frac{2}{7}$$

$$P(C_2|C) = \frac{\left(\frac{1}{3} \times \frac{8}{10}\right)}{\left(\frac{2}{3} \times \frac{3}{10}\right) + \left(\frac{1}{3} \times \frac{8}{10}\right)}$$

$$P(C_2|C) = \frac{4}{7}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

c)

—, —, —
 i. sum is even.

• ← E • E ✓

E E O X

G O E X

G O O ✓

O E E X

O E O ✓

O O E X

O O O X

20 +ve integer let $\{E\}$ be even
 let $\{O\}$ be odd

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

11, 12, 13, 14, 15, 16, 17, 18, 19, 20.

$$\left. \begin{aligned} n(E) &= 10 \\ P(E) &= \frac{2}{9} \\ n(O) &= 10 \\ P(O) &= \frac{1}{9} \end{aligned} \right\} \begin{aligned} C_1 &= 126 \\ C_2 &= 450 \\ P(20, 1E) &= \frac{15}{38} \end{aligned}$$

$$20 \quad C_3 = 1140$$

$$\frac{2}{19} + \frac{15}{38} = \frac{1}{2}$$

ii. Product is even.

6 6 6	✓
6 6 0	✓
6 0 6	✓
6 0 0	✓
0 6 6	✓
0 6 0	✓
0 0 6	✓
0 0 0	X

$$P(3 \text{ even}) = \frac{10C_3}{20C_3}$$

$$P(2 \text{ even}, 1 \text{ odd}) = \frac{10C_2 \times 10C_1}{20C_3}$$

$$P(1 \text{ even}, 2 \text{ odd}) = \frac{10C_1 \times 10C_2}{20C_3}$$

$$P(\text{Prod is even}) = \frac{17}{19}$$

OR.

$$P(3 \text{ odd}) = \frac{10C_3}{20C_3} = \frac{2}{19}$$

$$P(\text{Prod is even}) = 1 - \frac{2}{19} = \frac{17}{19}$$

Q3:-

a) A random variable is a variable that takes on numerical values based on the outcomes of a random phenomenon.

There are two main types of random variables:

Discrete Random Variables: Takes a countable no. of distinct value e.g. no. of heads in a series of coin tosses / no. of students in class.

Distribution function: Also known as cumulative distribution function that describes probability that a random variable X takes on a value less than or equal to a certain value x .

$$F_X(x) = P(X \leq x)$$

b) not done.

c)

$$f(x) = \begin{cases} Ax^3(1-x) & 0 \leq x \leq 1 \\ 0 & \text{elsewhere} \end{cases}$$

$$\int_0^1 Ax^3(1-x) dx$$

$$A \int_0^1 x^3 dx - \int_0^1 x^4 dx$$

$$A \int_0^1 x^3 - x^4 dx = 1$$

$$A \left[\int_0^1 x^3 dx - \int_0^1 x^4 dx \right] = 1$$

$$A \left[\frac{x^4}{4} \Big|_0^1 - \frac{x^5}{5} \Big|_0^1 \right] = 1$$

$$A \left(\frac{(1)^4}{4} - \frac{(1)^5}{5} \right) = 1$$

$$A \left(\frac{1}{4} - \frac{1}{5} \right) = 1$$

$$A \left(\frac{1}{20} \right) = 1$$

$$A = 20$$

$$\sigma^2 = \frac{10}{21} - \left(\frac{2}{3} \right)^2$$

$$= \frac{2}{63}$$

$$E(x) = \int_{-\infty}^{+\infty} x f(x) dx$$

$$E(x) = \int_0^1 x A(x^3)(1-x) dx$$

$$E(x) = A \int_0^1 x^4 - x^5 dx$$

$$E(x) = A \left[\frac{x^5}{5} \Big|_0^1 - \frac{x^6}{6} \Big|_0^1 \right]$$

$$E(x) = 20 \left(\frac{1}{5} - \frac{1}{6} \right)$$

$$E(x) = \frac{2}{3}$$

$$\text{Var} = \sigma^2 = E(x^2) - [E(x)]^2$$

$$E(x^2) = \int_0^1 x^2 A(x^3)(1-x) dx$$

$$E(x^2) = A \int_0^1 x^5 - x^6 dx$$

$$E(x^2) = A \left(\frac{1}{6} - \frac{1}{7} \right)$$

$$E(x^2) = 20 \left(\frac{1}{42} \right)$$

$$E(x^2) = \frac{10}{21}$$

